

Gael Berlanga Boemare

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Education

University of Cambridge

Cambridge, United Kingdom.

Master of Engineering, future specialisation in Information Engineering.

Sep. 2022 – May 2026

1st year grade: Upper Second-Class

Coursework: Mathematical Methods, Probability, Mechanics, Mechanical Vibrations, Thermodynamics, Structures, Material Science, Physical Principles of Electronics, Analysis of Circuits and Devices, Linear Circuits, Electromagnetics, Digital Circuits, Information Processing, Data Science and Computing.

Activities: Member of Cambridge University Algorithmic Trading Society and Downing College Rugby.

British School of Brussels

Brussels, Belgium

A-levels: A*, A*, A*

Sep. 2020 – Jul. 2022

Coursework: Mathematics, Physics, Further Mathematics - Decision 1 and Mechanics 1 modules.

Awards: European Statistics Competition 2020. Team of three, 1st place, 17 000 students, 16 countries.
Belgian Physics Olympiad 2021. 3rd place.

Technical Skills and Projects

Programming and Design: Python, R, C++, JavaScript, HTML, CSS, SolidWorks, AutoCAD.

Mars Lander Control System:

- Implemented orbital dynamics and numerical integrators in C++ to simulate a probe's orbit around Mars.
- Developed a proportional control system to successfully land the probe from orbit.

Experience

Engineering Intern, AGRO – OLEUM INGENIERIA S.L.

Summer 2023

- Collaborated with a team of engineers to aid in the production and development of the factory of San Miguel.
- Designed and edited factory plans using AutoCAD from data extracted from modelling software.
- Presented plans, led, and communicated to onsite builders.

Data Science Intern, Desbrest Institute of Epidemiology and Public Health

Winter 2023

- Developed algorithms for automated pulmonary health analysis from exhaled gases using R.
- Extracted meaningful data from curves of exhaled gases by fitting non-linear models.
- Performed principal component analyses followed by clustering algorithms to classify patients.
- Conducted a study linking coastal weather to stroke cases using R.
- Utilised multiple regression techniques, notably Lasso, to extract meaningful variables.

Languages and interests:

Languages: French and Spanish (native), English (professional fluency).

Interests: Weightlifting, Rugby, Recreational Programming.